

● HARD

● ADVANCED MATH

● ANSWER KEY

Equivalent expressions

01

10 answers · Rationale-rich · Teach from doc

Q1**C****C. 0**

Choice C is correct. Since the expression $x^2 + 6x - 7$ can be factored as $(x+7)(x-1)$, the given equation can be rewritten as $\frac{(x+7)(x-1)}{x+7} = ax + d$. Since $x \neq -7$, $x+7$ is also not equal to 0, so both the numerator and denominator of $\frac{(x+7)(x-1)}{x+7}$ can be divided by $x+7$. This gives $x-1 = ax + d$. Equating the coefficient of x on each side of the equation gives $a = 1$. Equating the constant terms gives $d = -1$. The sum is $1 + (-1) = 0$.

Choice A is incorrect and may result from incorrectly simplifying the equation. Choices B and D are incorrect. They are the values of d and a , respectively, not $a + d$.

Q2**.0177**

Accept also: .0178, 4/225

The correct answer is $\frac{4}{225}$. An expression of the form $\sqrt[k]{a}$, where k is an integer greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{1}{k}}$. Therefore, the given expression, where $n > 1$, is equivalent to $70n^{\frac{1}{5}}70n^{\frac{12}{6}}$. Applying properties of exponents, this expression can be rewritten as $70n^{\frac{1}{5}}70n^{\frac{1}{6} \cdot 2}$, or $70n^{\frac{1}{5}}70n^{\frac{1}{3}}$, which can be rewritten as $70n^{\frac{1}{5} + \frac{1}{3}}$, or $70n^{\frac{8}{15}}$. It's given that the expression $\sqrt[5]{70n} \sqrt[3]{70n}^2$ is equivalent to $70n^{30x}$, where $n > 1$. It follows that $70n^{\frac{8}{15}}$ is equivalent to $70n^{30x}$. Therefore, $\frac{8}{15} = 30x$. Dividing both sides of this equation by 30 yields $\frac{8}{450} = x$, or $\frac{4}{225} = x$. Thus, the value of x for which the given expression is equivalent to $70n^{30x}$, where $n > 1$, is $\frac{4}{225}$. Note that 4/225, .0177, .0178, 0.017, and 0.018 are examples of ways to enter a correct answer.

Q3**C**

C. $\frac{xy^2 + 13xy - 8y}{x^2y - 8xy}$

Choice C is correct. Factoring the denominator in the second term of the given expression gives $\frac{y+12}{x-8} + \frac{yx-8}{xyx-8}$. This expression can be rewritten with common denominators by multiplying the first term by $\frac{xy}{xy}$, giving $\frac{xyy+12}{xyx-8} + \frac{yx-8}{xyx-8}$. Adding these two terms yields $\frac{xyy+12+yx-8}{xyx-8}$. Using the distributive property to rewrite this expression gives $\frac{xy^2+12xy+xy-8y}{x^2y-8xy}$. Combining the like terms in the numerator of this expression gives $\frac{xy^2+13xy-8y}{x^2y-8xy}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

D

D. $\frac{x(x+3)}{x+1}$

Choice D is correct. Since $x^3 - 9x = x(x+3)(x-3)$ and $x^2 - 2x - 3 = (x+1)(x-3)$, the fraction $\frac{f(x)}{g(x)}$ can be written as $\frac{x(x+3)(x-3)}{(x+1)(x-3)}$. It is given that $x > 3$, so the common factor

$x-3$ is not equal to 0. Therefore, the fraction can be further simplified to $\frac{x(x+3)}{x+1}$.

Choice A is incorrect. The expression $\frac{1}{x+1}$ is not equivalent to $\frac{f(x)}{g(x)}$ because at $x=0$, $\frac{1}{x+1}$ has a value of 1 and $\frac{f(x)}{g(x)}$ has a value of 0.

Choice B is incorrect and results from omitting the factor x in the factorization of $f(x)$. Choice C is incorrect and may result from incorrectly factoring $g(x)$ as $(x+1)(x+3)$ instead of $(x+1)(x-3)$.

Q5**D**

D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

Choice D is correct. For $x > 1$ and $y > 1$, $x^{\frac{1}{3}}$ and $y^{\frac{1}{2}}$ are equivalent to $\sqrt[3]{x}$ and $\sqrt{y}$, respectively. Also, x^{-2} and y^{-1} are equivalent to $\frac{1}{x^2}$ and $\frac{1}{y}$, respectively. Therefore, the given expression can be rewritten as $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$.

Choices A, B, and C are incorrect because these choices are not equivalent to the given expression for $x > 1$ and $y > 1$.

For example, for $x = 2$ and $y = 2$, the value of the given expression is $2^{-\frac{5}{6}}$; the values of the choices, however, are $2^{-\frac{1}{3}}$, $2^{\frac{5}{6}}$, and 1, respectively.

Q6

C

C. 20

Choice C is correct. To express the sum of the two rational expressions on the left-hand side of the equation as the single rational expression on the right-hand side of the equation, the expressions on the left-hand side must have the same denominator. Multiplying the first

expression by $\frac{x+5}{x-5}$ results in $\frac{2(x+5)}{(x-2)(x+5)}$, and multiplying the second expression by

$\frac{x-2}{x-2}$ results in $\frac{3(x-2)}{(x-2)(x+5)}$, so the given equation can be rewritten as

$$\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}, \text{ or}$$

$$\frac{2x+10}{(x-2)(x+5)} + \frac{3x-6}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}.$$

Since the two rational expressions on the left-hand side of the equation have the same denominator as the rational expression on the right-hand side of the equation, it follows that $(2x+10) + (3x-6) = rx+t$. Combining like terms on the left-hand side yields $5x+4 = rx+t$, so it follows that $r=5$ and $t=4$. Therefore, the value of rt is $(5)(4) = 20$.

Choice A is incorrect and may result from an error when determining the sign of either r or t .

Choice B is incorrect and may result from not distributing the 2 and 3 to their respective terms

in $\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$. Choice D is incorrect and may result from

a calculation error.

Q7**C****C. 24**

Choice C is correct. If the equation is true for all x , then the expressions on both sides of the equation will be equivalent. Multiplying the polynomials on the left-hand side of the equation gives $5ax^3 - abx^2 + 4ax + 15x^2 - 3bx + 12$. On the right-hand side of the equation, the only x^2 -term is $-9x^2$. Since the expressions on both sides of the equation are equivalent, it follows that $-abx^2 + 15x^2 = -9x^2$, which can be rewritten as $(-ab + 15)x^2 = -9x^2$. Therefore, $-ab + 15 = -9$, which gives $ab = 24$.

Choice A is incorrect. If $ab = 18$, then the coefficient of x^2 on the left-hand side of the equation would be $-18 + 15 = -3$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side. Choice B is incorrect. If $ab = 20$, then the coefficient of x^2 on the left-hand side of the equation would be $-20 + 15 = -5$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side. Choice D is incorrect. If $ab = 40$, then the coefficient of x^2 on the left-hand side of the equation would be $-40 + 15 = -25$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side.

Q8

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Answer not provided in source data — see rationale below.

The correct answer is $\frac{7}{6}$. The value of $\frac{a}{b}$ can be found by first rewriting the left-hand side of the given equation as $\frac{x^{\frac{5}{2}}}{x^{\frac{4}{3}}}$. Using the properties of exponents, this expression can be rewritten as $x^{\left(\frac{5}{2} - \frac{4}{3}\right)}$. This expression can be rewritten by subtracting the fractions in the exponent, which yields $x^{\frac{7}{6}}$. Thus, $\frac{a}{b}$ is $\frac{7}{6}$. Note that $7/6$, 1.166, and 1.167 are examples of ways to enter a correct answer.

Q9

8

The correct answer is 8. Since each term of the given expression, $2x^3 + 42x^2 + 208x$, has a factor of $2x$, the expression can be rewritten as $2xx^2 + 2x21x + 2x104$, or $2xx^2 + 21x + 104$. Since the values 8 and 13 have a sum of 21 and a product of 104, the expression $x^2 + 21x + 104$ can be factored as $x + 8x + 13$. Therefore, the given expression can be factored as $2xx + 8x + 13$. It follows that the factors of the given expression are 2, x , $x + 8$, and $x + 13$. Of these factors, only $x + 8$ and $x + 13$ are of the form $x + b$, where b is a positive constant. Therefore, the possible values of b are 8 and 13. Thus, the smallest possible value of b is 8.

Q10

D

D. $\frac{9}{x + 14x - 5}$

Choice D is correct. The expression $\frac{4}{4x - 5} - \frac{1}{x + 1}$ can be rewritten as $\frac{4}{4x - 5} + \frac{-1}{x + 1}$. To add the two terms of this expression, the terms can be rewritten with a common denominator. Since $\frac{x + 1}{x + 1} = 1$, the expression $\frac{4}{4x - 5}$ can be rewritten as $\frac{x + 14}{x + 14x - 5}$. Since $\frac{4x - 5}{4x - 5} = 1$, the expression $\frac{-1}{x + 1}$ can be rewritten as $\frac{4x - 5 - 1}{4x - 5 - 1}$. Therefore, the expression $\frac{4}{4x - 5} + \frac{-1}{x + 1}$ can be rewritten as $\frac{x + 14}{x + 14x - 5} + \frac{4x - 5 - 1}{4x - 5 - 1}$, which is equivalent to $\frac{x + 14 + 4x - 5 - 1}{x + 14x - 5}$. Applying the distributive property to each term of the numerator yields $\frac{4x + 4 + -4x + 5}{x + 14x - 5}$, or $\frac{4x + -4x + 4 + 5}{x + 14x - 5}$. Adding like terms in the numerator yields $\frac{9}{x + 14x - 5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q8 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

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